Intracranial Hypertension & Herniation - Stroke

CLINICAL SIGNS OF HERNIATION

- **Motor decline:** Spontaneous GCS motor score decrease of ≥ 1 point.
- **Pupillary reactivity:** Decrease in pupillary reactivity (Neurological Pupil Index, NPi < 3).
- **Asymmetry:** New pupillary asymmetry or unilateral dilation (ipsilateral mydriasis).
- **Focal deficit:** New focal motor deficit or abnormal posturing (decorticate / decerebrate).
- **Cushing's Triad (Late Sign):** Systolic hypertension, bradycardia, and irregular respirations. ***Cushing triad is a LATE sign of brainstem compression.***

Syndrome	Anatomical Substrate	Clinical Exam & Diagnostic Trap
Uncal (Lateral)	Medial temporal lobe (uncus) pushed over tentorial edge	Ipsilateral sluggish/dilated pupil (CN III compressed), contralateral hemiparesis. **Kernohan's Notch** causes false-localizing ipsilateral hemiparesis.
Central (Axial)	Downward diencephalic and midbrain displacement	Progressive stupor, midpoint fixed pupils, decorticate to decerebrate posturing. Symmetrical signs often confused with metabolic encephalopathy.
Subfalcine (Cingulate)	Cingulate gyrus displaced under the falx cerebri	Often clinically silent, or presents with contralateral lower extremity weakness. **ACA compression** causes frontal/leg territory infarction.
Tonsillar (Downward)	Cerebellar tonsils forced through the foramen magnum	Cushing's triad, flaccid quadriplegia, respiratory arrest.

MANAGEMENT

GENERAL APPROACH

Fundamental Measures

- Elevate HOB 30°; strict neutral midline neck alignment to preserve venous outflow.
- Euvolemia (isotonic saline; avoid hypotonic D5W).
- Temperature < 38.0°C.
- Normocapnia (pCO_2 35–45 mmHg).
- When ICP is monitored, CPP = MAP - ICP; many protocols target CPP around >60 mmHg, individualized to disease context.

Medical Interventions

- **Analgesia/sedation (fentanyl/propofol):** Target RASS -1 to +1 to prevent coughing, agitation, or ventilator dyssynchrony.
- **Mannitol 20%:** 1 g/kg IV bolus over 20–30 min. Must use in-line filter. Hold if Osm > 320 or Gap ≥ 20.
- **Hypertonic Saline (HTS):** 3% (150–250 mL bolus) or 23.4% (30 mL rescue bolus; central access only). Hold if Na > 155 or Cl > 115.
- **Ventilation:** Maintain normocapnia (PaCO2 35–45 mmHg). For impending herniation only, use brief controlled hyperventilation targeting about 30–35 mmHg while definitive therapy is initiated; avoid prophylactic or prolonged hypocapnia.
- **Refractory ICP Elevation:** High-dose barbiturate therapy (pentobarbital) titrated to burst suppression on EEG.

CSF DIVERSION

EVD placement for acute hydrocephalus, intraventricular hemorrhage (IVH), or mass effect with ventriculomegaly.

DECOMPRESSIVE SURGERY

Malignant MCA (DHC): Age ≤ 60 years, clinical decline, infarct ≥ 50% MCA territory, within 48h (DECIMAL/DESTINY).

Cerebellar Stroke: Suboccipital craniectomy for brainstem compression, 4th ventricle effacement, or hydrocephalus.

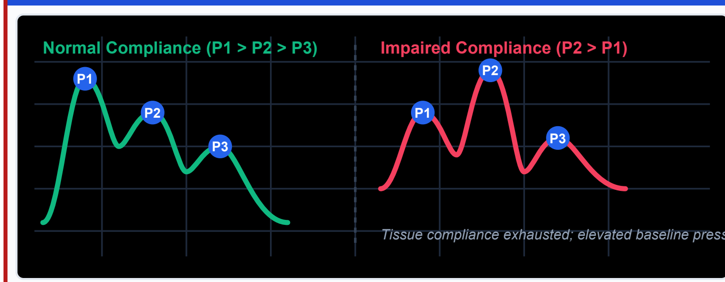
Intracranial Hemorrhage (ICH): Cerebellar ICH with deterioration, brainstem compression, hydrocephalus, or large size requires urgent surgical evaluation. Supratentorial/lobar ICH evacuation or decompression is case-dependent rather than routine.

Management is not necessarily sequential

For active herniation or rapid clinical/radiographic deterioration, immediately initiate medical interventions & call Neurosurgery.

Corticosteroids are not indicated for cytotoxic edema in stroke and increase infection risk.

ICP WAVEFORM ANALYSIS



- **P1 (Percussion wave):** Arterial pulsation.
- **P2 (Tidal wave):** State of intracranial compliance (elastic reserve).
- **P3 (Dicrotic wave):** Venous pulsations.
- **Compliance States:**
  - **Normal Compliance:** P1 > P2 > P3 (elastic brain tissue easily cushions pulsations).
  - **Impaired Compliance / High ICP:** P2 > P1 (brain tissue reserve exhausted; high risk of herniation).

CPP = MAP - ICP

In patients with intracranial hypertension or mass effect, cerebral perfusion is highly pressure dependent, and cautious BP lowering is recommended.